Aufgabe 1:

Gesucht ist die kleinste Teilmenge χ der Potenzmenge von $\{0,1,2,3,4\}$, welche folgende Eigenschaften besitzt:

1. $\{0,2,4\} \in \chi$
2. $A \in \chi \implies \{0,1,2,3,4\} \setminus A \in \chi$
3. $A, B \in \chi \implies A \cup B \in \chi$

$\{0,2,4\} \in \chi$

$\overset{2.}{\implies} \{0,1,2,3,4\} \setminus \{0,2,4\} = \{1,3\} \in \chi$

$\overset{2.}{\implies} \{0,1,2,3,4\} \setminus \{1,3\} = \{0,2,4\} \in \chi$

$\overset{3.}{\implies} \{0,2,4\} \cup \{1,3\} = \{1,3\} \cup \{0,2,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{2.}{\implies} \{0,1,2,3,4\} \setminus \{0,1,2,3,4\} = \emptyset \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \{0,2,4\} = \{0,2,4\} \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \{1,3\} = \{1,3\} \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{2.}{\implies} \{0,1,2,3,4\} \setminus \emptyset = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \{0,2,4\} \cup \emptyset = \emptyset \cup \{0,2,4\} = \{0,2,4\} \in \chi$

$\overset{3.}{\implies} \{1,3\} \cup \emptyset = \emptyset \cup \{1,3\} = \{1,3\} \in \chi$

$\overset{3.}{\implies} \{0,1,2,3,4\} \cup \emptyset = \emptyset \cup \{0,1,2,3,4\} = \{0,1,2,3,4\} \in \chi$

$\overset{3.}{\implies} \emptyset \cup \emptyset = \emptyset \in \chi$

Endergebnis: $\chi = \{\emptyset, \{0,2,4\}, \{1,3\}, \{0,1,2,3,4\}\}$